Understanding SIR models, one flow at a time

Beginner mathematical and statistical reading notes
Prepared for Md. Ikramul Haq Khan — MS thesis literature review
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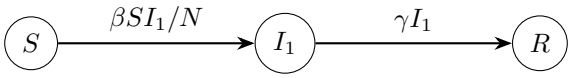
Reading context. Companion to Fudolig and Howard (2020), Figure 1 and the "Before emergence" discussion, especially the passage on birth/death, vaccination, transmission and removal coefficients. This note also consolidates the earlier compartment explanation in this conversation. These are learning notes, not an approved thesis specification or a claim of new results.

1 People, groups, and time

Main idea. A compartment is a group of people with the same modeled disease status.

Symbol	Meaning at time t
$S(t)$	Susceptible: can catch the original strain
$I_1(t)$	Infectious with strain 1 (the original strain)
$R(t)$	Recovered and protected against strain 1
$V(t)$	Vaccinated/protected against strain 1
N	Total population: $S + I_1 + R + V$

The subscript 1 labels the strain; it does not mean one infected person. The paper also includes naturally immune people in V. Here we explain the vaccination mechanism shown in its figure.



This simplified path shows infection and recovery. Births, vaccination and deaths are added below. Each arrow label is a *flow rate*, measured in people per unit of time.

2 A dictionary of the parameters

Symbol	Interpretation	Units
μ	Per-person natural death rate; also the birth coefficient	time^{-1}
p	Fraction of newborns vaccinated, $0 \le p \le 1$	no units
β	Transmission coefficient	time^{-1}
γ	Recovery/removal coefficient from infection	time^{-1}

Distinguish state from parameter: $I_1(t)$ changes along a trajectory; a fixed γ describes the rule governing recovery. A coefficient is not itself the total flow. Although the paper calls p a vaccination rate, it functions here as a proportion of births.

3 Why births and deaths use μ

Main idea. Multiply a per-person rate by a group size to obtain a total flow. Births enter at rate μN. Of these, fraction p enter V and fraction $1-p$ enter S:

$$\text{births into } S = (1-p)\mu N, \qquad \text{births into } V = p\mu N.$$

Natural deaths leave each compartment at rates μS, μI_1, μR and μV. For example, μS has units $(\text{day}^{-1})(\text{people}) = \text{people/day}$ if days are the time unit. If total births are 10 per unit time and $p = 0.7$, 7 enter V and 3 enter S. These are illustrative numbers, not estimates from the paper. **Model assumption:** newborns receive vaccination at entry. There is no adult $S \to V$ transition in Figure 1. Vaccination fully protects against strain 1, with no waning in this baseline.

4 Why infection has flow $\beta S I_1/N$

Main idea. More infectious people increase exposure; more susceptible people create more opportunities for new infection.

$$\lambda(t) = \beta \frac{I_1(t)}{N}, \qquad \text{total infection flow} = \lambda(t) S(t).$$

Here I_1/N is the infectious fraction. The quantity λ is the infection *hazard*: the instantaneous infection rate per susceptible person. Under homogeneous mixing, contacts encounter infectious people according to their population fraction. One interpretation is $\beta = cq$, with c contacts per person per unit time and q transmission probability per relevant contact. **Standard incidence:** with c fixed, doubling population size at the same infectious fraction leaves each susceptible person's hazard unchanged. Total infections double if S doubles. This differs from a density-dependent term $\widetilde{\beta} S I_1$, whose coefficient has different units. **Worked example.** Let $N = 1000$, $S = 800$, $I_1 = 50$, and $\beta = 0.4$ per day. Then

$$\lambda = 0.4 \frac{50}{1000} = 0.02 \text{ per day}, \qquad \lambda S = 0.02(800) = 16 \text{ people/day}.$$

This is the instantaneous flow, not a guarantee of exactly 16 infections tomorrow. If $S = 0$ or $I_1 = 0$, the infection flow is zero.

5 Why recovery has flow γI_1

Each infectious person has recovery hazard γ, so the total recovery flow is γI_1. If $\gamma = 0.2$ per day and $I_1 = 50$, it is 10 people/day. With constant recovery hazard and no competing exit, infectious duration is exponential with mean $1/\gamma = 5$ days in the corresponding Markov interpretation. With natural death also possible, time until either recovery or death has mean $1/(\gamma + \mu)$. The ODE itself tracks continuous group totals, not individual waiting times.

6 Translate arrows into differential equations

Main idea. For every group: rate of change equals total inflow minus total outflow. For a general compartment $C(t)$,

$$\frac{dC}{dt} = \text{inflow} - \text{outflow}.$$

The derivative dC/dt describes the slope right now. For a small interval h, $C(t+h) \approx C(t) + h\,dC/dt$. The approximation gets better as h shrinks when the solution is smooth. Reading all arrows gives

$$\begin{aligned}
\frac{dS}{dt} &= (1-p)\mu N - \frac{\beta S I_1}{N} - \mu S, \\
\frac{dI_1}{dt} &= \frac{\beta S I_1}{N} - \gamma I_1 - \mu I_1, \\
\frac{dR}{dt} &= \gamma I_1 - \mu R, \\
\frac{dV}{dt} &= p\mu N - \mu V.
\end{aligned}$$

Read the infectious equation aloud: the infectious group gains new infections and loses recoveries and natural deaths. With the previous example and $\mu = 0.001$ per day, its net change is $16 - 10 - 0.05 = 5.95$ people/day. Fractional values are legitimate in this continuous approximation. **Conservation check.** Adding the four equations cancels internal transfers:

$$\frac{dN}{dt} = \mu N - \mu(S + I_1 + R + V) = 0.$$

Thus total population stays fixed because births balance deaths. Individual compartments can still change. Here a closed population means no migration; births and deaths are explicitly allowed.

7 When does infection increase?

Factoring the infectious equation gives

$$\frac{dI_1}{dt} = I_1 \left(\beta \frac{S}{N} - (\gamma + \mu) \right).$$

For $I_1 > 0$, infection increases when $\beta S/N > \gamma + \mu$. Intuitively, each infectious person generates infections faster than people leave the infectious group. Define the basic reproduction number for this baseline by

$$\mathcal{R}_0 = \frac{\beta}{\gamma + \mu}.$$

It combines transmission with mean time before recovery/death in a fully susceptible population. The current effective value is $\mathcal{R}_{\text{eff}}(t) = \mathcal{R}_0 S(t)/N$. At the disease-free equilibrium, $S/N = 1 - p$, so $\mathcal{R}_v = \mathcal{R}_0(1-p)$. The threshold concerns growth near that state; it does not guarantee persistence in every random simulation.

8 Statistical meaning: rates are not probabilities

Main idea. A rate describes how quickly events occur; a probability describes the chance of an event over an interval. For a constant isolated hazard a, the probability of its event within time h is $1 - e^{-ah}$, approximately ah for small h. If $\lambda = 0.02$ per day and exposure stays constant with no competing exits, the one-day infection probability is $1-e^{-0.02} \approx 0.0198$. A hazard need not be below 1; a probability must be. **A possible stochastic extension.** At the current state, an infection event changes (S, I_1, R, V) by $(-1, +1, 0, 0)$, with event rate $a_{\text{inf}} = \beta S I_1/N$. A recovery changes it by $(0, -1, +1, 0)$, with rate $a_{\text{rec}} = \gamma I_1$. For a sufficiently small h, the probability of one specified event type is $a_k h + o(h)$. This is a local approximation, not an unrestricted time-step formula. These random events generate different trajectories even with identical parameters. Nonlinear mean dynamics generally do not close exactly: $E[SI_1] \neq E[S]E[I_1]$ in general. Therefore the deterministic curve need not equal the exact stochastic mean. **Population-size caution.** Independent random birth/death events allow the total population to fluctuate even though their deterministic drifts balance. An exactly fixed-N stochastic model requires a compatible replacement mechanism and adviser confirmation. This note does not choose the thesis event mechanism.

9 Connect to infection curves and distribution fitting

Prevalence $I_1(t)$ counts people currently infectious. Incidence flow $j(t) = \beta S(t) I_1(t)/N$ counts new infections per time. They have different areas:

$$A_T = \int_0^T I_1(t)\,dt \quad \text{(infectious person-time)}, \qquad C_T = \int_0^T j(t)\,dt \quad \text{(new infections)}.$$

For time in days, 10 infectious people for 3 days contribute 30 person-days. This does not identify the number of distinct infected people. Finite horizon T must be specified; persistent positive prevalence makes infinite-horizon prevalence area diverge. In repeated stochastic runs, $A_T^{(1)}, \ldots, A_T^{(B)}$ form a sample of epidemic burdens. A distribution fit describes their variation across runs. Fitting a curve over time is a separate task; successive time points are not independent repeated epidemics. Confirm the target, initial state, horizon and parameter settings with the adviser before fitting.

10 Reference and learning checks

Fudolig, M., and Howard, R. (2020). The local stability of a modified multi-strain SIR model for emerging viral strains. *PLOS ONE*, 15(12), e0243408. https://doi.org/10.1371/journal.pone.0243408.
Baseline definitions follow this source; worked numbers and statistical explanations are instructional additions.

Check your understanding: Why does infection appear with opposite signs in S and I_1? Why is p unitless? Why does constant N not imply constant I_1? What changes if the curve is incidence rather than prevalence? Explain each in words before manipulating equations.